

Appendix

Dataset

Pre-training Dataset We employ two widely-used datasets for pre-training: Enwiki8 and WikiText-103.

Enwiki8 is a character-level dataset containing the first 100MB of English Wikipedia text. It is commonly used for evaluating compression and character prediction performance, with bits per character (bpc) as the metric—lower bpc indicates better compression. Its compact and highly compressible nature makes it ideal for testing lightweight models under limited-context settings.

WikiText-103 is a large-scale word-level corpus derived from high-quality Wikipedia articles, preserving punctuation and document structure. It is suitable for modeling long-range dependencies, with perplexity (ppl) as the evaluation metric—lower ppl implies stronger language modeling ability.

Finetune dataset For the fine-tuning stage, we adopt four representative GLUE benchmark datasets—SST-2, QQP, QNLI, and RTE—for downstream evaluation, which span sentiment classification, paraphrase detection, and natural language inference tasks.

- **SST-2:** The Stanford Sentiment Treebank II (SST-2) is a binary sentiment classification task on single sentences from movie reviews. With approximately 67K training examples, models predict whether each sentence is positive or negative. Performance is assessed via accuracy, reflecting sentiment prediction correctness.
- **QQP:** Quora Question Pairs (QQP) is a paraphrase detection task that determines if two Quora questions are semantically equivalent. It includes roughly 364K training pairs. Evaluation uses both accuracy, capturing correctness and class balance.
- **QNLI:** Question-answering Natural Language Inference (QNLI) converts SQuAD-style question-sentence pairs into a binary entailment task: does the sentence contain the answer? It provides around 105K training pairs, and uses accuracy as its evaluation metric.
- **RTE:** The Recognizing Textual Entailment (RTE) task is a binary natural language inference dataset consolidated from multiple RTE challenges. It is relatively small, with about 2.5K training examples, and evaluated via accuracy to judge whether one sentence entails the other.

Experiment Configuration

Pre-training configuration During the pre-training stage, we perform dataset-specific pretraining based on the Transformer-XL architecture. The training is conducted using 4 NVIDIA A800 GPUs. Table 1 summarizes the detailed configurations used in the main pretraining experiments.

Finetune configuration During the fine-tuning stage, we further adapt the pretrained model to downstream tasks using four GLUE datasets: SST-2, QQP, QNLI, and RTE. All fine-tuning experiments are conducted using 4 NVIDIA A800 GPUs. Table 2 summarizes the detailed configurations used in the fine-tuning stage.

Layer	Optimizer	Lr	Batch Size	Steps	Expert Num
8	Adam	0.0007	48	80k	16

Table 1: Experimental configurations for pre-training on Enwiki8 and Wikitext103.

Layer	Optimizer	Lr	Batch Size	Steps	Expert Num
8	Adam	0.0001	48	4k	16

Table 2: Experimental configurations for fine-tune on four downstream datasets.

Baseline models

We select the most relevant state-of-the-art MoE baseline models for comparison in our main experiments. Below is a brief introduction to each baseline:

- **SMOE:** The standard Sparse Mixture-of-Experts model leverages learned softmax-based routing to dynamically assign tokens to a subset of experts, supporting scalable computation and conditional capacity allocation.
- **SMOE-Dropout:** A plug-and-play routing scheme that freezes a randomly initialized router and progressively increases the Top- k expert selection during training, yielding a self-slimmable architecture with flexible expert activation and robust performance across various routing budgets.
- **HyperRouter:** Combines a fixed hypernetwork and trainable layer embeddings to generate layer-specific routing parameters, achieving a dynamic and semantically adaptive routing strategy with improved stability across layers.
- **RMOE:** Utilizes a GRU-based memory module to model expert activation trajectories across layers as sequential patterns, enabling enhanced information flow and promoting semantic alignment in routing decisions.

Algorithm Framework

To address the limitations of single-layer routing decisions in sparse MOE models, we propose MCF-MoE, a unified routing framework that incorporates Multi-layer Context Fusion to enhance expert selection. The detailed algorithmic framework is summarized in Algorithm 1.

Performance under Varying Top- K Expert Activations

To systematically investigate the impact of varying the number of activated experts K on model performance, we conducted a series of comparative experiments on Enwiki8 and Wikitext103. The experimental results are summarized in Table 3. Overall, as K increases, most routing strategies exhibit performance fluctuations or even degradation. In contrast, the proposed MCF-MoE demonstrates remarkable stability and consistently superior performance.

K	Enwiki8 (bpc)				Wikitext103 (ppl)			
	SMOE-Drop	HyperRouter	RMOE	MCF-MoE	SMOE-Drop	HyperRouter	RMOE	MCF-MoE
1	1.144	1.848	1.652	1.133	27.678	79.253	51.020	28.012
2	1.141	1.262	1.481	1.126	26.405	39.557	41.193	25.975
4	1.141	1.179	1.494	1.125	26.541	32.704	42.533	26.087
8	1.140	1.155	1.530	1.125	27.128	30.710	46.399	26.918
16	1.140	1.149	1.604	1.125	28.053	30.722	54.036	28.012

Table 3: Performance comparison of different MoE models on Enwiki8 (measured in bpc) and Wikitext103 (measured in ppl) under varying top- K settings. Lower values indicate better performance.

Algorithm 1: MCF-MoE: Multi-layer Context Fusion MOE

Input: Current layer input $X_l \in \mathbb{R}^{B \times T \times D}$, layer index l
Parameter: Number of experts E , top- K selection, local radius r , local top- k k_{local}
Output: Expert indices I and routing weights W

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1: Initialize historical context  $\mathcal{H} = \{H_0, H_1, \dots, H_{l-1}\}$ 
2: if  $l = 0$  or  $|\mathcal{H}| = 0$  then
3:   logits = Linear( $X_l$ )
4: else
5:   // Cross-layer Context Module
6:    $H_{concat} = \text{Concat}(\mathcal{H})$ 
7:    $C_{cross} = \text{CrossAttention}(X_l, H_{concat})$ 
8:   // Local Similarity Module
9:    $S = X_l \cdot X_l^T$ 
10:   $M = \text{CreateLocalMask}(S, r)$ 
11:  for each token  $i$  do
12:    Select  $k_{local}$  neighbors via masked  $S$ 
13:     $C_{local}[i] = \sum_j \text{weight}_{ij} \cdot X_l[j]$ 
14:  end for
15:  // Context Fusion and Expert Routing
16:   $X_{fused} = C_{cross} + C_{local} + \text{LayerEmbedding}(l)$ 
17:  logits = ExpertProjection( $X_{fused}$ )
18: end if
19:  $I, W = \text{TopK}(\text{softmax}(\text{logits}/\tau), K)$ 
20: Update  $\mathcal{H} \leftarrow \mathcal{H} \cup \{X_l\}$ 
21: return  $I, W$ 

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On Enwiki8, conventional methods such as RMOE and SMOE-Drop are highly sensitive to changes in K , showing noticeable non-monotonic trends and significant degradation at larger K values. In comparison, MCF-MoE maintains near-constant performance across different K settings, consistently operating within an optimal range, indicating strong robustness to varying activation sizes. Similarly, on Wikitext103, most baseline methods suffer from performance instability or decline as K increases. In contrast, MCF-MoE consistently achieves lower perplexity across various K configurations, outperforming other strategies and exhibiting superior generalization capabilities. This suggests that MCF-MoE is more effective in aligning local and global semantics in complex tasks such as language modeling, thereby enhancing expert consistency across tokens.

These findings indicate that MCF-MoE exhibits stronger adaptability and robustness under different Top- K configu-

rations. By alleviating the instability and semantic mismatch issues prevalent in existing routing strategies, MCF-MoE provides a more reliable solution for sparse expert models operating under dynamic computational constraints.

Performance Comparison Across Model Scales

To investigate the performance of different gating mechanisms across varying model scales, we trained each method on the Enwiki8 dataset using Transformer-XL under both large and small configurations for 80K steps. As shown in Figure 1, MCF-MoE consistently achieves the best performance across both scales, indicating its superior modeling capacity and compression efficiency. In contrast, RMOE performs the worst in both settings, which may be attributed to its reliance on low-dimensional hidden states and limited ability to capture fine-grained contextual dependencies. Besides, SMOE-Dropout shows relatively consistent performance across scales, suggesting that its stochastic expert deactivation strategy enhances model robustness, though it remains less effective in capturing precise semantic information.

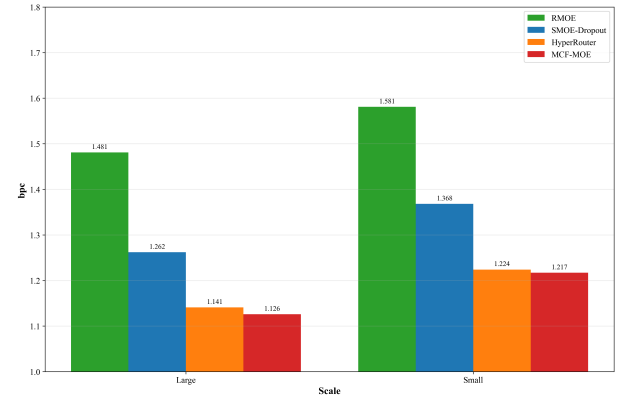


Figure 1: Performance of different gating mechanisms under two Transformer-XL scales on the Enwiki8 dataset. Lower is better.

Expert Collaboration and Specialization Analysis

To further investigate the expert collaboration patterns under different gating mechanisms, we evaluate the expert co-occurrence similarity on the Enwiki8 test set during inference. This metric measures the probability that two ex-

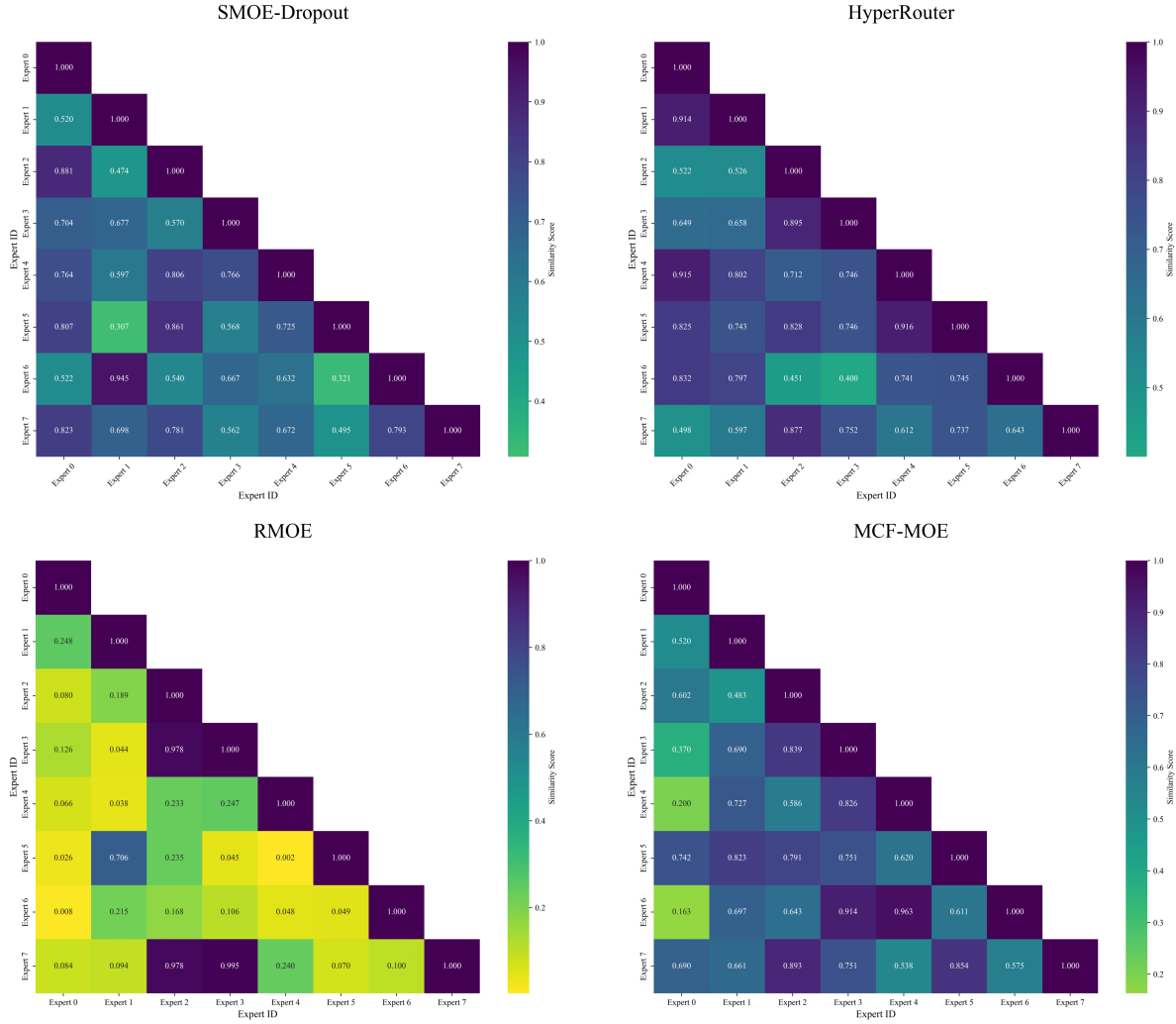


Figure 2: Expert Occurrence Similarity Matrix of MCF-MoE and other models. The heatmap illustrates the co-activation probability between expert pairs when processing the same token. Higher values indicate stronger collaboration tendencies.

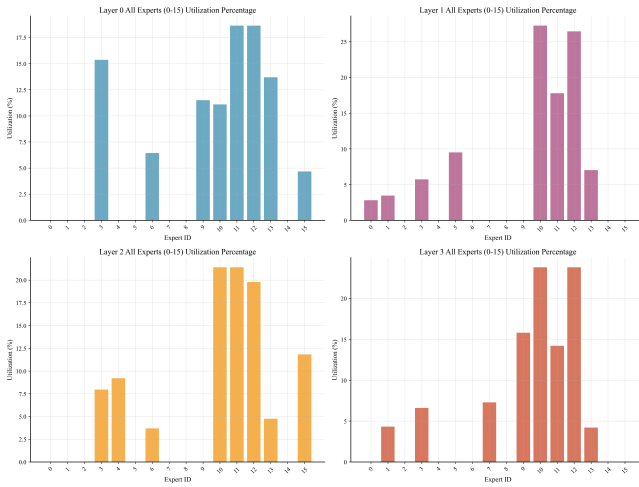
perts are simultaneously activated when processing the same token. Moderate co-occurrence is beneficial for functional complementarity, while overly high or low co-occurrence may lead to expert redundancy or isolation, respectively.

As shown in Figure 2, MCF-MoE exhibits a more structured and balanced co-occurrence distribution, maintaining moderate co-activation among semantically correlated expert pairs. This pattern indicates a clear division of labor and cooperation. The results validate that the multi-constraint filtering strategy employed in MCF-MoE—such as distance-aware suppression and semantic similarity regulation—effectively guides expert collaboration, enhancing expressive capacity while preserving functional diversity. SMOE-Dropout demonstrates a slightly higher level of co-occurrence, with certain expert pairs showing abnormally high co-activation. This suggests that although its random expert dropout strategy improves robustness, it lacks fine-grained control over expert specialization and division. Hy-

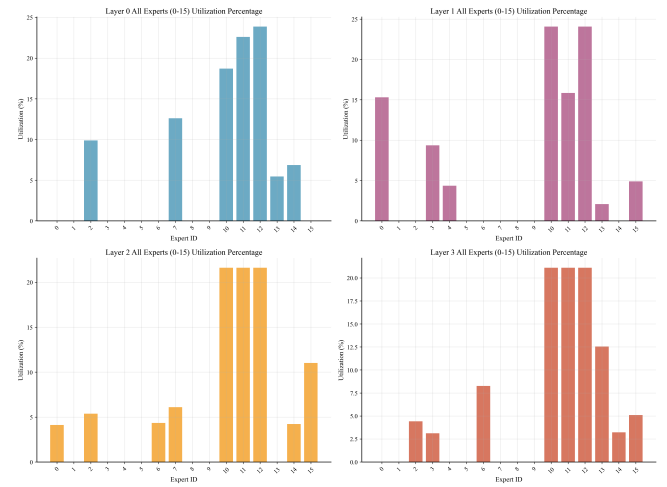
perRouter shows generally high co-occurrence across most expert pairs, reflecting strong routing consistency. However, the excessive overlap in expert responsibilities may hinder full utilization of the model capacity. RMOE presents an extremely sparse co-occurrence pattern, where only a few expert pairs exhibit high co-activation while most remain nearly isolated. This indicates that RMOE’s routing mechanism is overly binding on certain semantic branches, leading to poor collaboration and underutilization of experts. In summary, MCF-MoE constructs a more reasonable routing topology, striking a better balance between cooperation and specialization. It provides a promising direction toward enhancing the scalability and generalization of sparse expert models.

Routing Efficiency Analysis

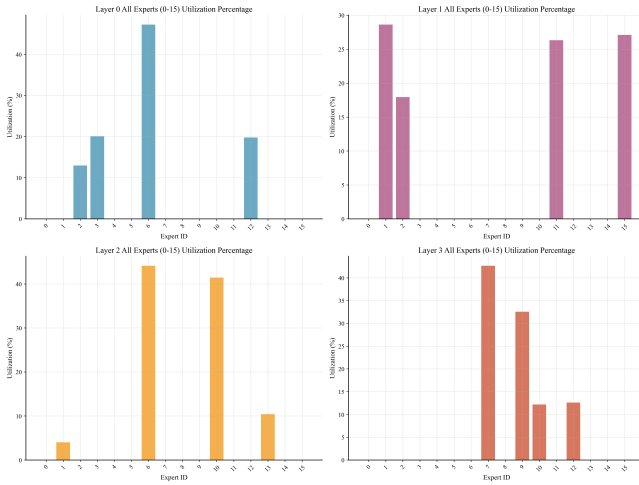
To evaluate the routing efficiency of different sparse expert strategies, we compare their throughput, measured in



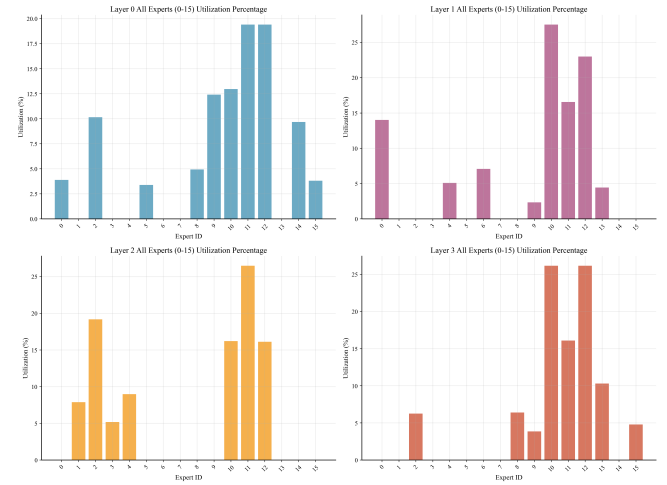
(a) SMOE-Dropout



(b) HyperRouter



(c) RMOE



(d) MCF-MOE

Figure 3: Comparison of expert utilization across all layers in four routing methods.

Algorithm	Throughput (K tokens/s)
SMOE-Drop	382.42
HyperRouter	381.78
RMOE	380.44
MCF-MoE	383.52

Table 4: Throughput (K tokens/s) comparison across different routing strategies. A higher throughput indicates greater execution efficiency in processing tokens per second.

thousands of tokens processed per second (K tokens/s). A higher throughput signifies more efficient expert activation and reduced routing overhead, which is particularly vital for scaling models in real-world applications. As shown in Table 4, MCF-MoE achieves the highest throughput among all compared methods, indicating that MCF-MoE not only preserves model performance but also improves computa-

tional efficiency. Compared to conventional methods, whose routing mechanisms often involve stochastic expert deactivation (e.g., SMOE-Drop) or sequential memory modeling with additional overhead (e.g., RMoE), MCF-MoE adopts a lightweight, multi-constraint filtering scheme that streamlines token-to-expert assignment, which effectively reduces redundant computations during token-to-expert assignment.

Hierarchical Expert Utilization Analysis

Figure 3 presents the expert utilization across all layers for four models. SMOE-Dropout, HyperRouter, and MCF-MoE exhibit a relatively diverse expert activation pattern, with many different experts being engaged across layers, and a few (e.g., Expert 10–12) consistently demonstrating high usage. In contrast, RMoE activates significantly fewer experts per layer, resulting in a large number of experts that are rarely or never selected. This limited activation not only constrains the model’s ability to fully exploit the expert pool

but also reflects inefficient resource utilization, which may hinder both generalization and robustness.

Future Work

In this work, we propose MCF-MoE, a novel sparse expert routing framework that integrates global cross-layer context and local similarity-aware context to enhance expert selection and cooperation. Extensive experiments across multiple benchmarks demonstrate the effectiveness and generalizability of our approach.

For future work, we aim to further investigate the information flow dynamics among functionally complementary experts, with the goal of deepening our understanding and optimization of inter-expert communication and coordination. Specifically, we plan to (1) model the latent dependency structures between experts, and (2) enable adaptive expert specialization driven by downstream task signals. Furthermore, we will explore extending MCF-MoE to broader application domains such as multimodal learning, where efficient and dynamic expert collaboration is essential.